teuscher Lab
teuscher-lab.com

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[redacted]

- codefest/cf01/profiling/resnet18_analysis.md (top-5 MAC table + AI calculation)
- project/heilmeier_draft.md (first 3 questions answered)
- project/algorithm_diagram.png (or .pdf)
- Repo must be public and accessible by Sunday 11:59 pm.

Project — Heilmeier questions

1. Draft answers to the first 3 Heilmeier questions (https://www.darpa.mil/about/heilmeier-catechism) for your project idea. Save as project/heilmeier_draft.md.
2. Draw a high-level block diagram of your target algorithm (hand-drawn photo or use any tool, e.g., your LLM or PyFlowchart).
3. Save as project/algorithm_diagram.png (or .pdf).

CMAN

1. first layer has 784 nodes. Each node feeds into each of the subsequent 256 nodes. Since we know that each node in the first layer has 256 connections.

So,
for first layer

$$\#\text{Mac operations} = 784 \text{ nodes} \cdot \frac{256 \text{ connections}}{\text{node}}$$
$$= 200{,}704$$

∴ second layer

$$\# = 256 \times 128 = 32{,}768$$

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∴ third = $128 \times 10 = 1280$

5. $784 + 256 + 128 + 10 = 1178$ "neurons"

and we need 1 activation per neuron. So 1178 activations.

$1178 \cdot 4$ Bytes $= 4712$ Bytes

6. $(2 \times 234{,}752) / (4712 + 939{,}008)$

$469{,}504 / 943{,}720$

$= 0.498$

2. total MACs for forward pass:

$200{,}704 + 32{,}768 + 1280$

$= 234{,}752$ per forward pass

3. We know now that we have 234,752 Mac operations, for each one of those operations, we can assume there is a weight per operation.

So, 234,752 weights are utilized.

4. So, we have 234,752 weights and 4 Bytes per weight. So total memory is 939,008 bytes

CamScanner